PREDICTING BIOLOGICAL AGE FROM CELL-FREE RNA SEQUENCING

My end-of-studies internship in a nutshell

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CONTENT

My project uses **cfRNA sequencing** of healthy individuals to predict **biological age**, combining **machine learning models** (Elastic Net, XGBoost) with **co-expression network analysis** (WGCNA) towards an interpretable biological clock.

These slides describe the process I followed to answer my research question.

01. **Theoretical context** *p 3 to 6*

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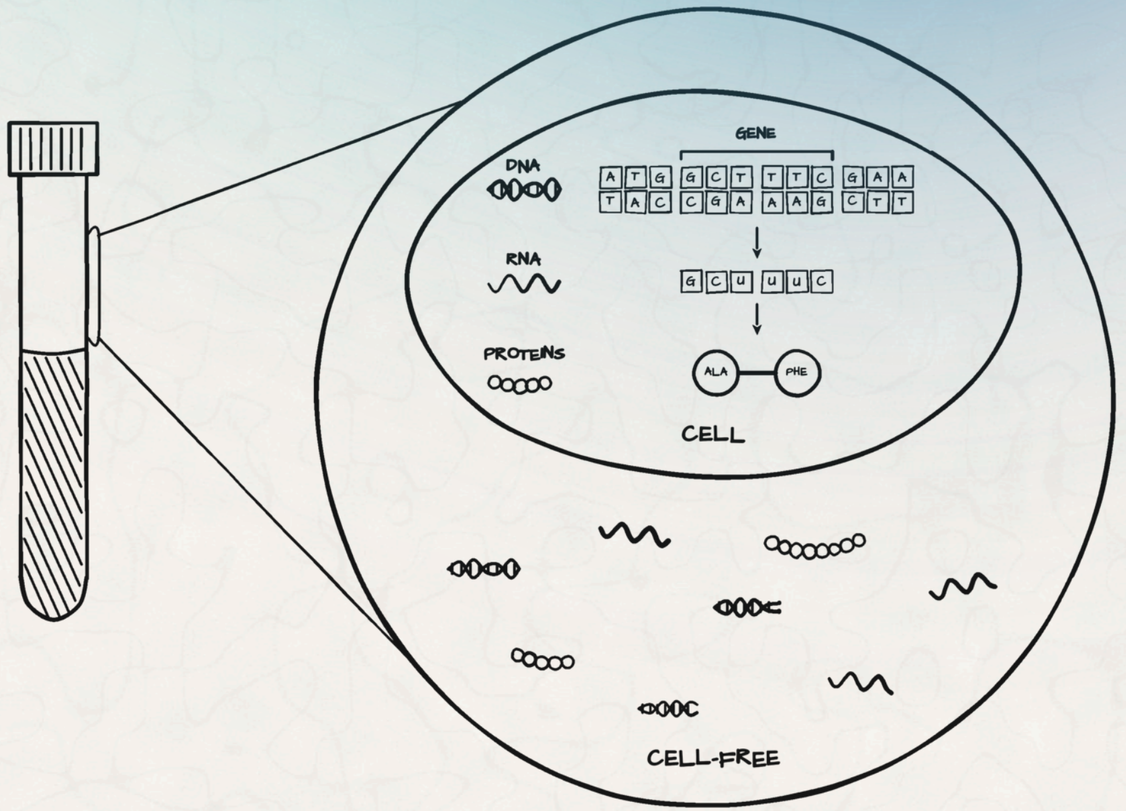
WHAT IS CELL-FREE RNA ?

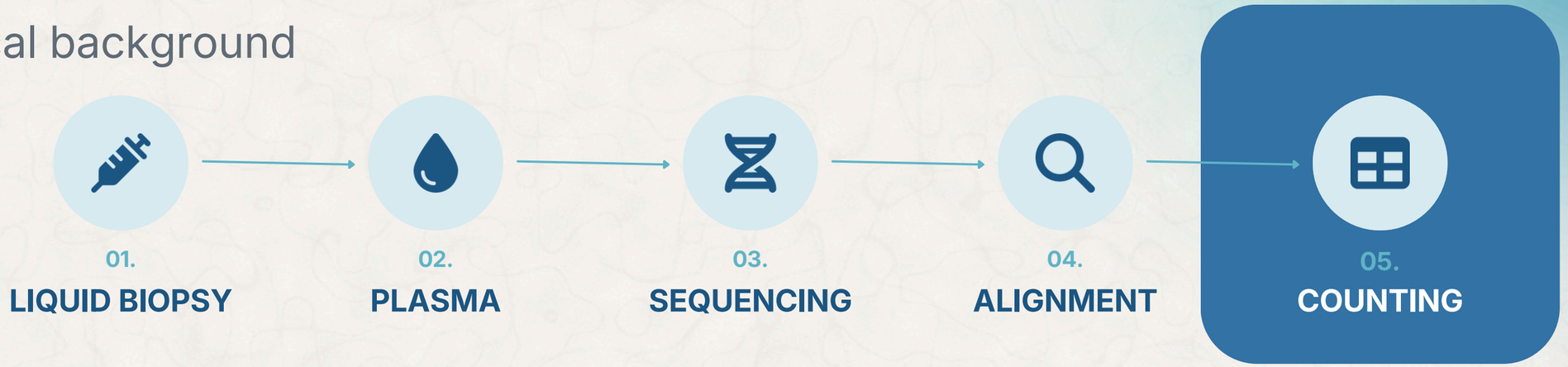
Theoretical background

- **DNA is** the **cell's genetic instruction manual,** stored in the nucleus. It is **organised into genes**, each one the **recipe for a protein.**
- **RNA is the working copy**: to make a protein, the cell first copies an active gene into messenger RNA, which is then translated into protein.
- **RNA therefore reflects gene activity**: which genes a cell is using, and how much, at a given moment.
- Most RNA stays inside cells, but some is **released into the blood plasma**: this is **cell-free RNA** (cfRNA).
- Because it comes from cells all over the body, cfRNA gives a **snapshot of the whole organism's state** from a single blood draw, not just of one tissue.



HOW TO GET THE GENE ACTIVITY ?

Theoretical background



- After a blood draw, the **plasma is isolated** and its **cell-free RNA is sequenced** into millions of short fragments called reads. Each **read is aligned to the reference genome**, and the **reads falling on each gene are counted**.
- The **more reads a gene has, the more active it is**.
- Some genes are ageing markers: their countshould rise or fall with age.
- For example, if gene B is involved in inflammation, which increases with age, older individuals should have more reads for gene B.

Age	Sample Id	Gene A	Gene B	Gene C
80	Patient A	1	**20**	2
35	Patient B	1	6	3
40	Patient C	2	5	2

The gene expression matrix!

But what if we could...

based on RNA data

Machine learning models

BUILD TRANSCRIPTOMIC CLOCKS THAT COULD CAPTURE AGEING ?

WHAT IS AGEING ?

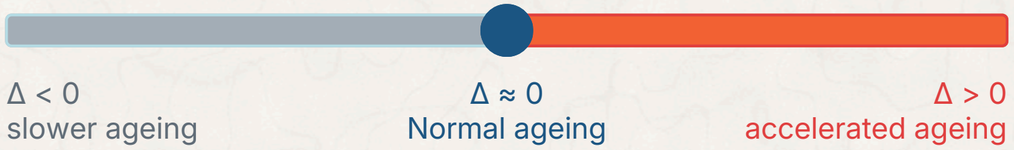
Calculating biological age

- **Throughout life, our cells accumulate damage** (DNA errors, telomere shortening ...). Repair mechanisms weaken over time, so this damage builds up and tissues gradually lose their function.

- However, **not everyone accumulates damage at the same pace**. Researchers introduced the notion of **biological age**, which reflects the body's actual state of damage.

BIOLOGICAL AGE = CHRONOLOGICAL AGE +Δ

Where chronological age is the number of days lived by the organism and delta (Δ) is the reflection of the functional state and damage accumulation of your cells.



Our main hypothesis is that for **healthy** subjects, Δ is 0 and their **biological age equals their chronological age**

INTERNSHIP FRAMEWORK

Building a clock

My research question

Can a machine learning model detect ageing in cfRNA and turn my compagny's data into a transcriptomic clock?

My challenges

The potential of building a clock from cfRNA is huge but the challenge is great.
The RNA fragments are known to be fragile and the slightest difference in how a sample is handled can leave its mark. The field is so young that no cfRNA clock has been built yet. However, transcriptomic clocks already succeed on tissue and blood RNA and cfRNA itself is already used to detect diseases, proof that it carries rich biological information.

My tasks

1 IDENTIFY AGEING SIGNAL IN cfRNA

2 TRAIN ML MODELS

MATERIALS

Technical Infrastructure and Tools

- All of the work was carried out on virtual machines within the Flomics cloud infrastructure hosted on **Amazon Web Services**.
- The core data manipulation and modeling work was performed on **Jupyter Notebooks**, using a pipeline developed primarily in **Python** for data manipulation and predictive modeling. Dependency and environment management relied on the **Conda** ecosystem, utilizing the standard data science suite: **pandas, NumPy, scikit-learn**
- Gene co-expression network analyses were performed separately, directly in **R.**

SAMPLE COLLECTION

About the main metadata

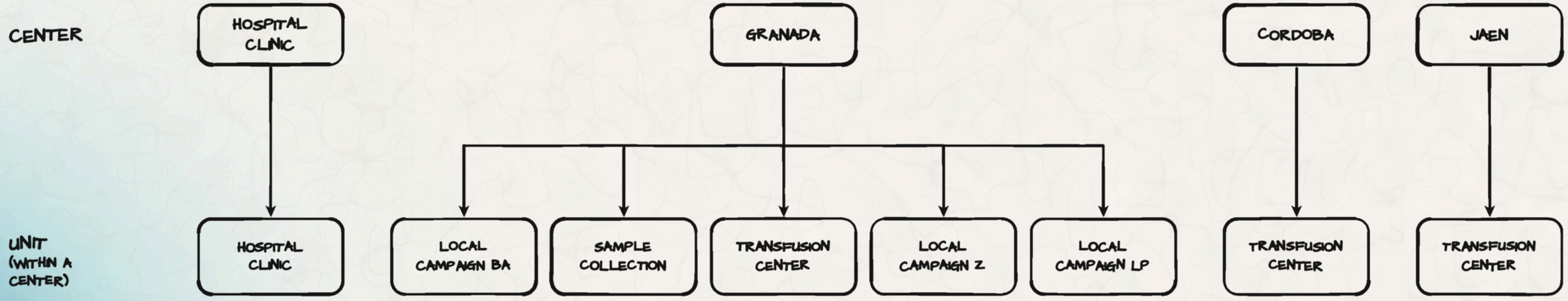
The data come from the cohort described in **A Robust Cell-Free RNA Approach for the Early Detection of Colorectal Cancer** (Monteagudo-Mesas et al.). Developed by the **Flomics team on a cohort sequenced in its own wet lab**, this study shows that cell-free RNA can detect early-stage colorectal cancer.

collection_date
Date of the blood draw

collection_center
Hospital in charge of collection

collection_health_care_unit_within_center
Unit within the centre

All samples were collected in **four Spanish hospitals**: Barcelona (Hospital Clínic), Granada, Córdoba and Jaén.



SUBJECTS

About the main metadata

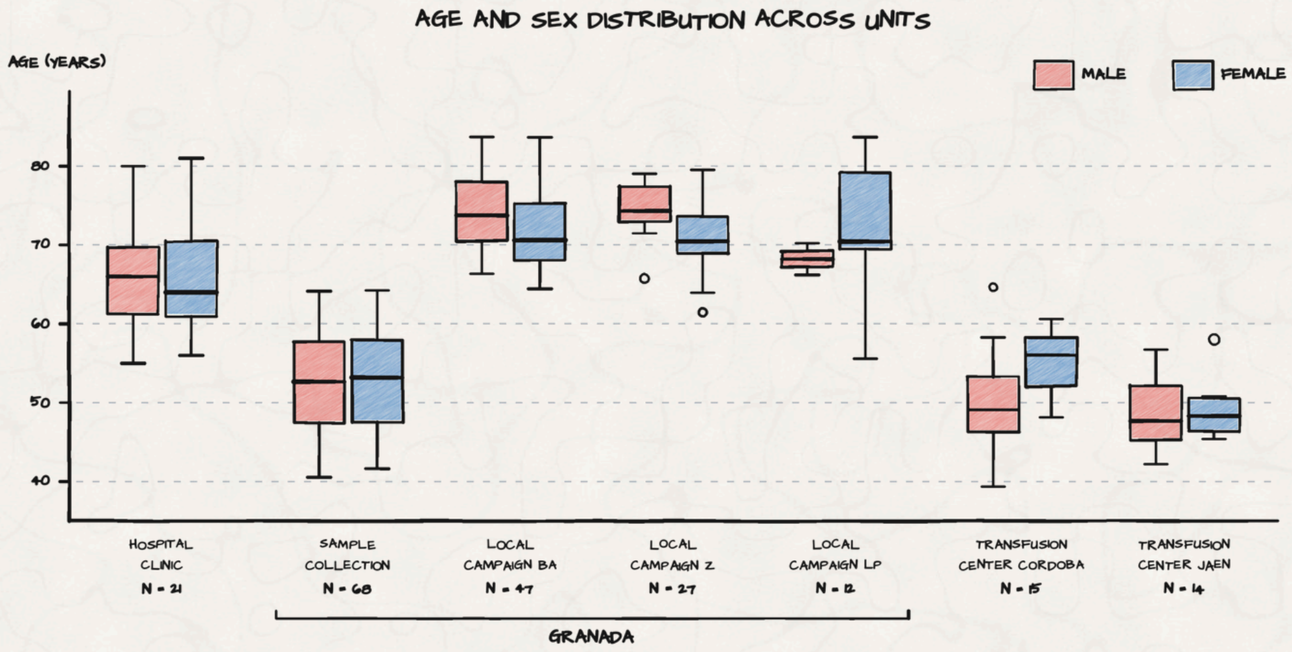
Each sample comes with these three key patient variables. The analysis is limited to **healthy donors, so that biological age can be compared with chronological age**.

Flomics built this dataset to study cancer. **“Healthy” means free of cancer and non-cancer dicease**, but other conditions are not always documented, so health status remains partly uncertain.

Healthy people rarely have blood drawn, so their samples are **scarcer** and come from different collection campaigns. Those real-world constraints lead to imbalances in age and sex between units. This led to a **correlation of 0.85 between age and units**.

But because the **number of samples is limited, these imbalances must be handled rather than removed**.

age
chronological age at blood draw
sex
biological sex
health_status
*cancer disease/non cancer disease/**healthy***



QUALITY CONTROL

About the main metadata

Flomics applies **strict quality control** at every step. Plasma is double-spun and checked for haemolysis, old samples are excluded, and DNA is removed during RNA extraction. Each sequenced sample must pass thresholds on exonic reads, spliced reads and library diversity. Even with these checks, residual platelets remain one of the main sources of variation in cell-free RNA.

platelet
platelet fraction predicted by deconvolution

Platelets are a major source of variation, as shown by Nesselbush et al. in ***An ultrasensitive method for detection of cell-free RNA*.** What they found :

- Residual platelets left after plasma isolation are a key variable: higher centrifugation speeds reduce both cfRNA concentration and platelet gene expression.
- Platelets are the single largest source of expression variation in cfRNA, together with natural differences in platelet counts between individuals.

GENE EXPRESSION MATRIX

What the model actually uses

14,000 genes × 204 patients

sample_id	Gene A	Gene B	Gene C	⋯
Patient_A	1.103	3.581	2.736	⋯
Patient_B	0.374	4.012	3.733	⋯
⋮	⋮	⋮	⋮	⋱

With so many variables and few samples, a model can fit the training data perfectly by memorising noise, leaving the model unable to generalise to new samples. It requires **constrained models and careful feature selection**.

This expression matrix captures far more than ageing-related variation. Gene expression in cfRNA reflects a complex mixture of biological and technical signals. **Biological variation includes factors such as sex, physiological state, recent illness, medication use, and lifestyle, many of which are not systematically recorded in our dataset**. For healthy donors, age and sex are the main available demographic variables. In contrast, technical sources of variation are generally better documented. Consequently, the **ageing signal**, which is **weak and distributed across many genes**, can be difficult to distinguish from these other sources of variability.

EXPLORING THE TECHNICAL SIGNAL

Correlation between samples

what we want

Age	Center
80	Granada
78	Jaen
25	Granada

Sample Id	Gene A	Gene B	Gene C
Patient A	**1**	**20**	**6**
Patient B	1	18	5
Patient C	8	2	6

what we see

Age	Center
80	Granada
25	Granada
78	Jaen

Sample Id	Gene A	Gene B	Gene C
Patient A	**1**	**20**	**6**
Patient B	1	18	5
Patient C	8	2	6

Samples from the **same hospital** (Granada) are **correlated together**. Correlation between samples is driven more by batch effect than by real biological factors (including age).

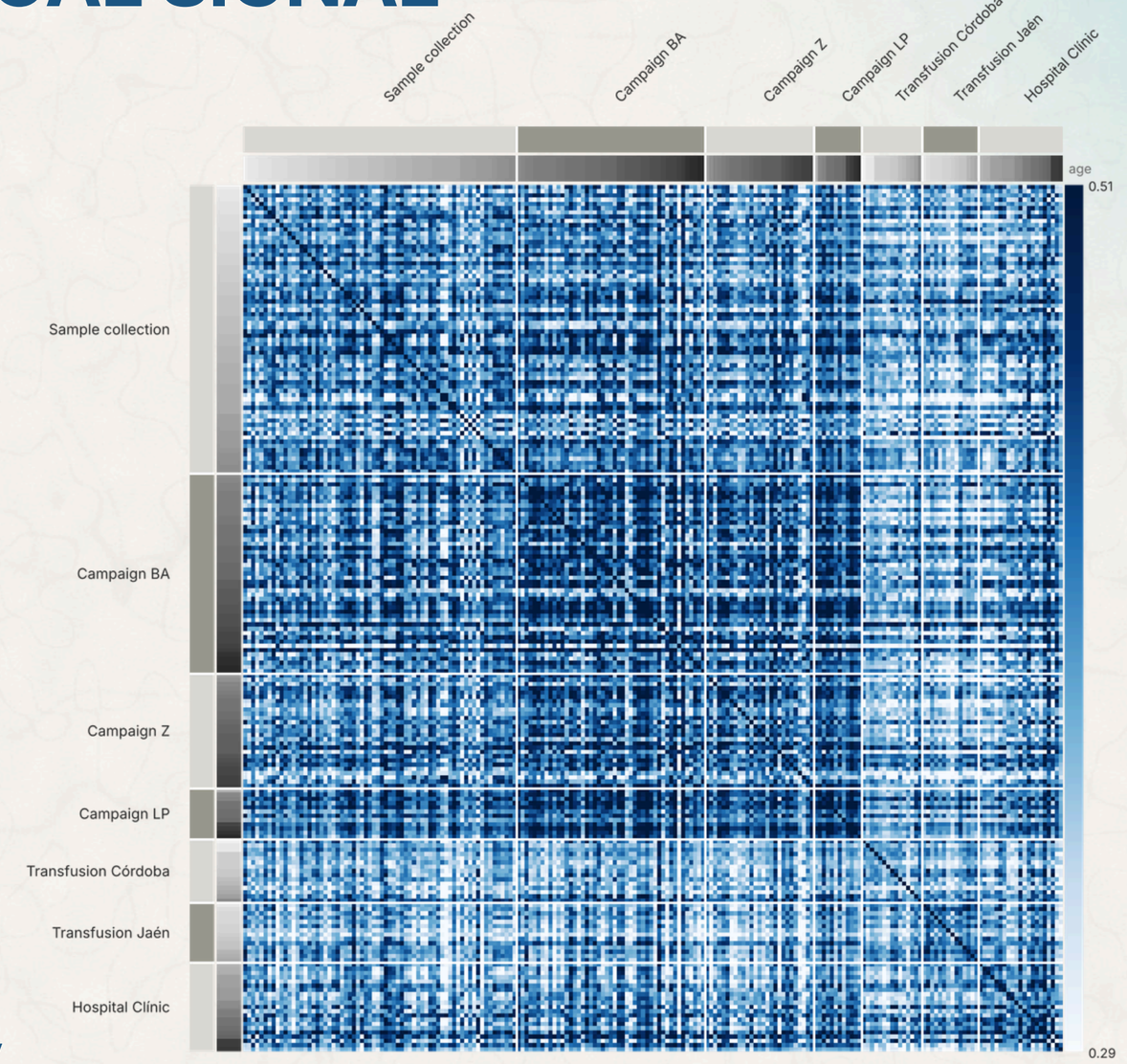
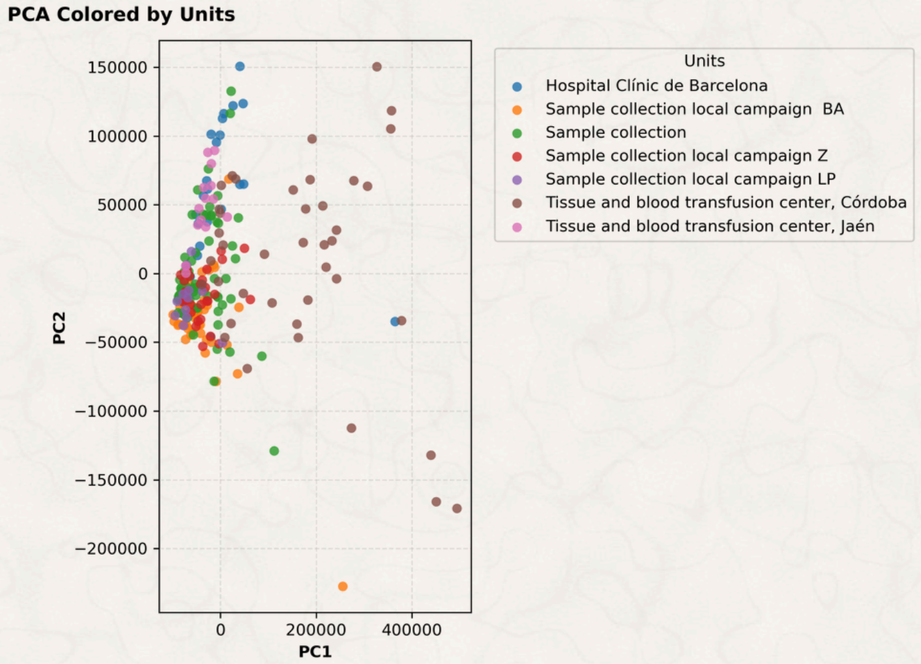
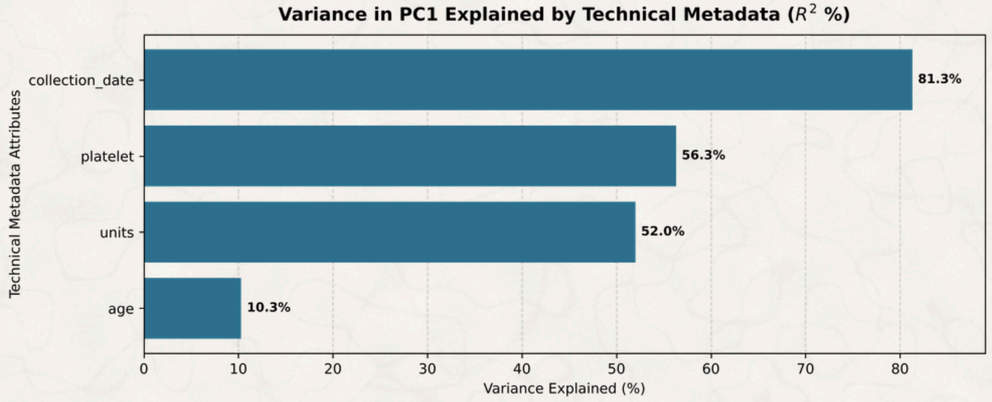


Fig: Heatmap of the Pearson correlation between samples, grouped by unit, units grouped by hospital

EXPLORING THE TECHNICAL SIGNAL

What explains the first principal components ?





The first two principal components capture the directions of greatest variance in the data: PC1 the largest, and PC2 the largest remaining direction uncorrelated with PC1.

When samples are coloured by unit, they form distinct clusters, most clearly for Córdoba. This suggests that much of the **variation between samples is associated with the collection unit, we are facing batch effect**. Quantifying how much of the variance of PC1 each covariate explains (R^2) confirms this: **technical covariates** such as collection date (81%), platelet contamination (56%) and unit (52%) **explain far more of PC1 than age** (10 %).

BATCH EFFECT CORRECTION : RUVG

Remove Unwanted Variation, guided by platelet genes

Remove Unwanted Variation using negative control genes (RUVg) is a method used in transcriptomics to remove unwanted sources of variation from gene expression data while preserving the biological signal of interest. RUVg model is :

$$\log Y = X\beta + \mathbf{W\alpha} + \varepsilon$$ *with Y: gene expression, X: variable of interest (age), W: unwanted factor, β and α: coefficients and ε: noise*

Reflects platelet contamination (we want to remove it)

The hypothesis (studied in the previously mentioned *An ultrasensitive method for detection of cell-free RNA article*) is that platelet genes do not depend on age: their variation reflects noise. We can use **platelet specific genes as control genes**.

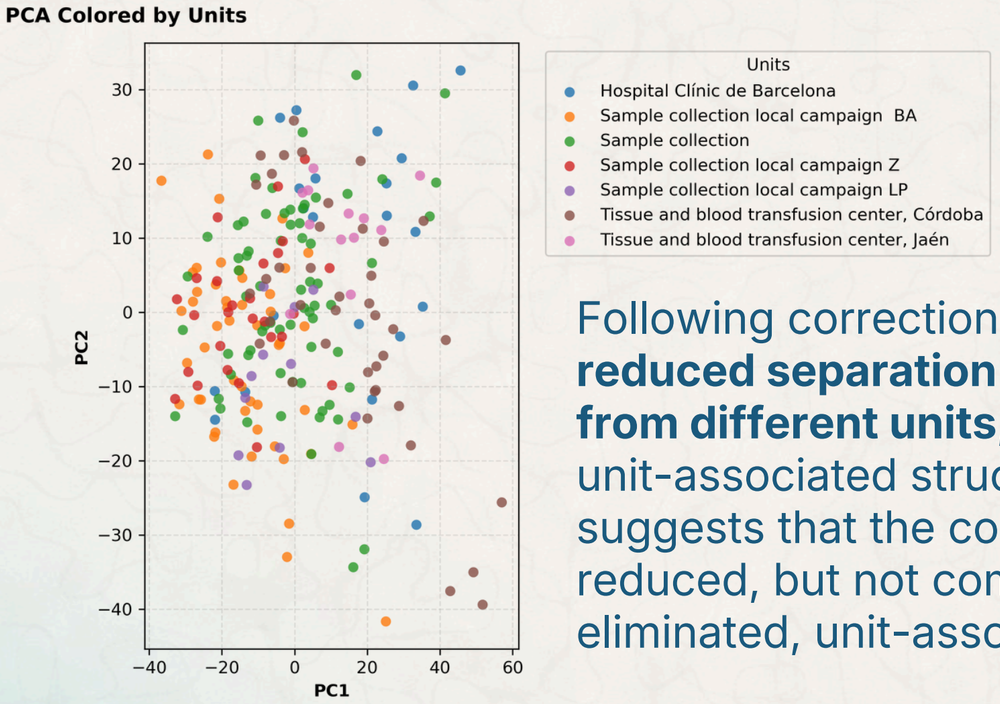
We select **platelet-specific genes** and calculate a contamination score by performing a PCA to identify the direction in which the common variation is strongest among these platelet genes. **PC1 assigns a score to each sample,** these scores form the **vector W**, representing the estimated contamination level.

Then, to measure the sensitivity of each gene, we **perform a regression of each gene against W**. The slope **α** indicates the extent to which that gene tracks the contamination.

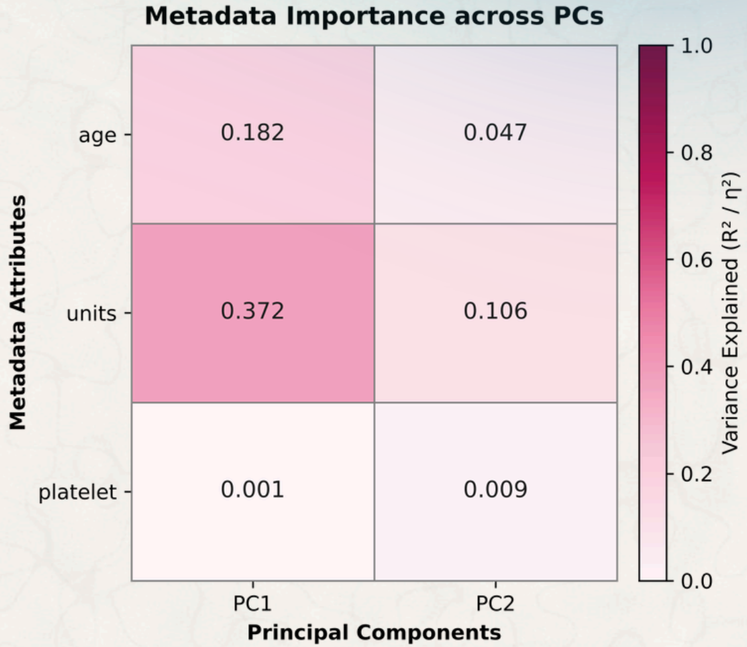
Finally we **subtract the term Wα**.

BATCH EFFECT CORRECTION : RUVG

Results



Following correction, the **PCA shows reduced separation between samples from different units**, although some unit-associated structure remains. This suggests that the correction has reduced, but not completely eliminated, unit-associated variation.



The metadata contribution to PC1 and PC2 further supports this observation: **although platelet-related variation appears to have been mostly removed, the effect of the originating unit remains detectable**. The underlying source of this residual variation is difficult to identify based on the current analysis. We will therefore explore a more direct correction strategy using limma to further assess and mitigate unit-associated effects.

BATCH EFFECT CORRECTION : LIMMA

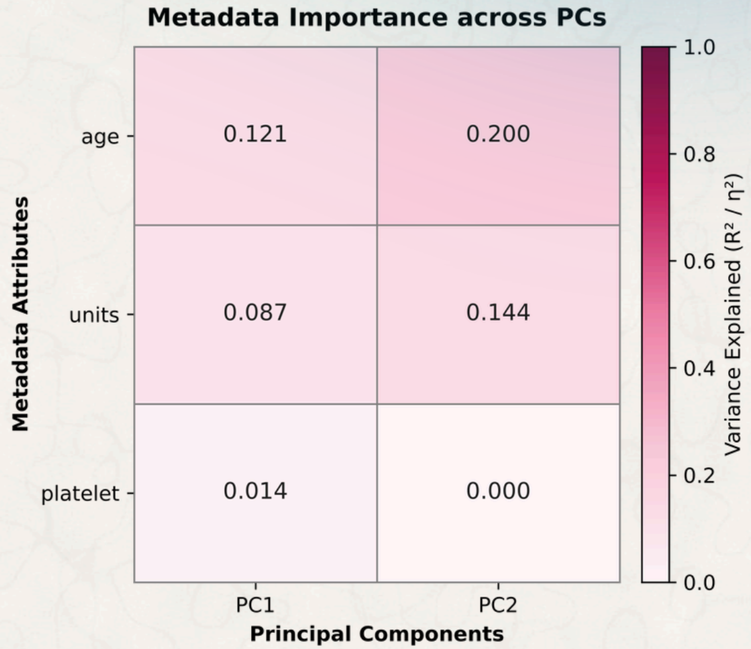
Principle and results

The limma approach uses a **linear modelling framework** to estimate and adjust for known sources of variation in gene expression. Initially, the unit-associated variation could be partly attributed to platelet contamination, but following correction, residual differences between units remain.

$$\log Y = X\beta + Z\gamma + \varepsilon$$

Reflects the unit effect

with Y: gene expression, X: variable of interest (age), Z: unwanted factor due to unit, γ and β: coefficients and ε: noise



After the correction, unit-related variation appears to have been reduced, and age signal preserved.

MODEL SELECTION : ELASTIC NET REGRESSION

A Linear model

Elastic Net predicts **age as a weighted sum of gene expressions, then penalise the weights** with a Lasso penalty (pushes the weights of weakly useful genes to exactly 0) and a Ridge penalty (shrinks all weights towards 0 without ever cancelling them) so the model cannot memorise noise.

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- **Balance**: The Lasso penalty limits overfitting, while the Ridge penalty distributes the weight. and genes work as a team ...
- **Interpretability**: Each selected gene has a weight with a sign and a magnitude reflecting gene/age relationship

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- **Linear only**: It does not account for threshold effects or gene interactions.

MODEL SELECTION : XGBOOST

A Tree-based model

XGBoost (eXtreme Gradient Boosting) builds a **family of decision trees**. Each tree asks **yes/no questions about gene expression** (“Is gene C > 7?”) **to group patients who have similar prediction errors**. It then applies the same correction to each group, and the next tree corrects the remaining errors.

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- **Nonlinear:** It captures threshold effects (one question = one threshold) and gene interactions (two consecutive questions).

Feature selection idea: select genes with a Lasso, then train XGBoost on them, to combine the strengths of both models.

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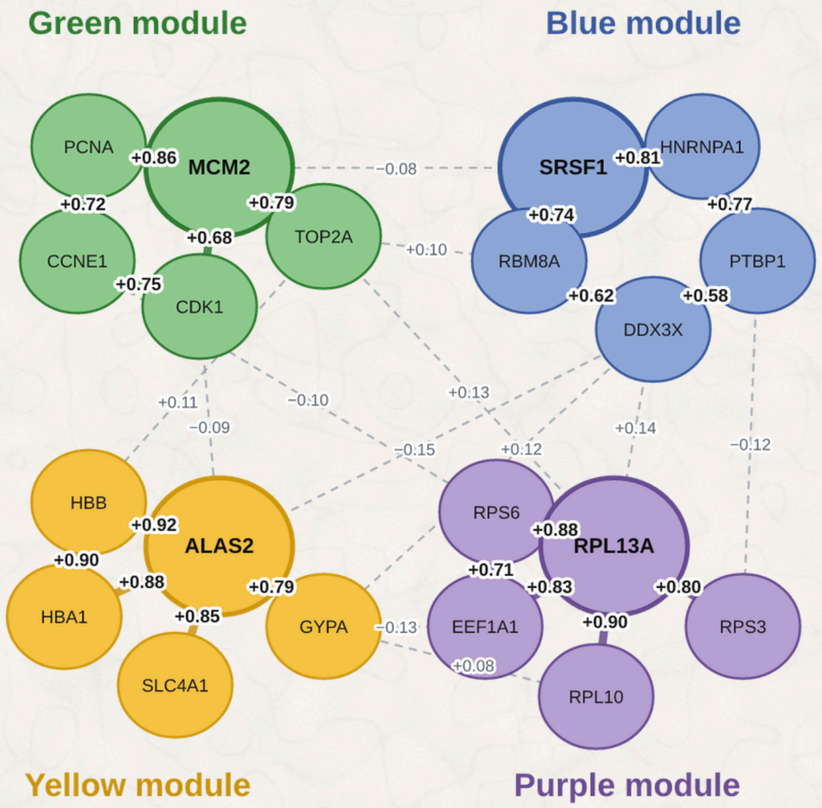
- **Sensitive to batch effects:** a single question about a gene with a shift is enough to isolate a center.
- **No extrapolation:** a prediction is always an average of training patients.
- **Less interpretable:** we know which genes are used, but not in what direction they vary with age.

FEATURE SELECTION : WGCNA

Gene co-expression network



Before selecting the genes to retain, we first sought to **characterize the underlying structure of the gene-expression data**. Ideally, gene selection should be guided by biological relevance, prioritizing genes with known functional roles and a plausible association with ageing.

Weighted Gene Co-expression Network Analysis looks at genes as a network. Each gene is a node, and edges are correlation, reflecting the co-expression. This gave **11 modules of 116 to 1,257 genes**, while 71% of genes stayed unassigned.

FEATURE SELECTION : WGCNA

Which module could be informative ?

To assess the biological relevance of the identified gene modules, we performed **enrichment analyses using two complementary databases**: **Gene Ontology** (GO), which provides information on gene functions and biological processes, and the **Kyoto Encyclopedia of Genes and Genomes** (KEGG), which describes functional pathways. A function or pathway was considered enriched when a module contained more genes associated with it than expected by chance.

We then **selected the most informative module** based on several criteria:

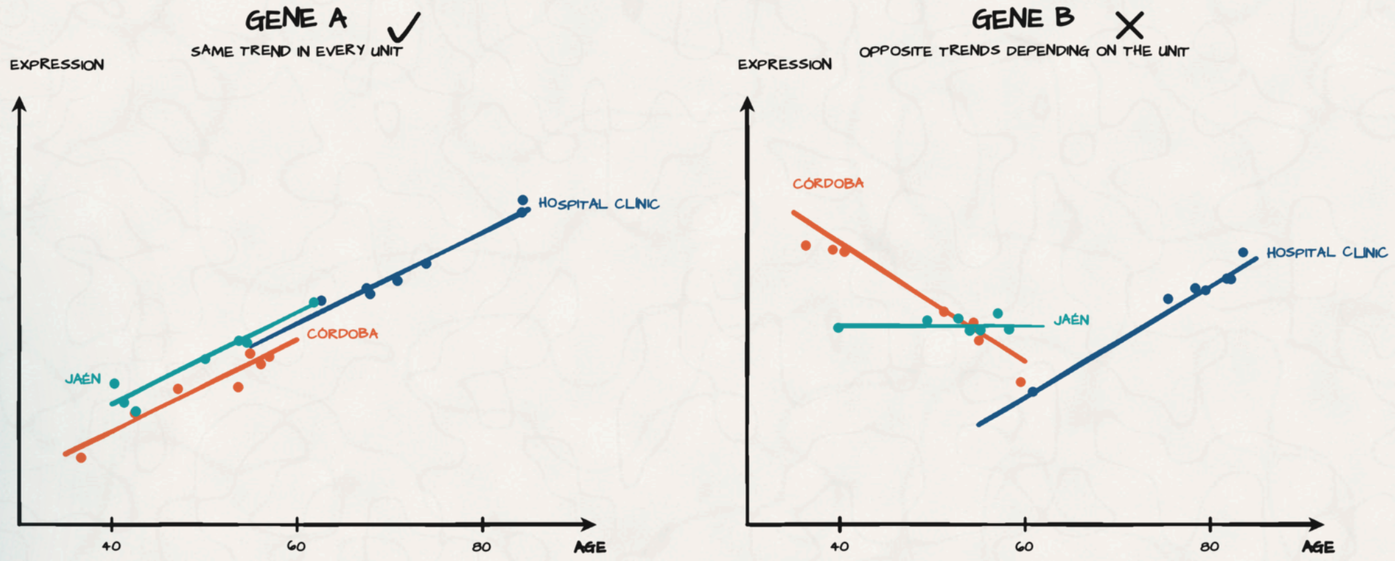
- A **strong correlation** between the module's first principal component (PC1) and age
- A **low adjusted enrichment p-value**
- A **low number of distinct KEGG pathway** groups sharing no genes, and **high coverage**, defined as the proportion of annotated module genes belonging to at least one enriched pathway.
- We considered whether the **enriched functions were consistent with established biological knowledge of ageing**.

Based on these criteria, the **red module** (271 genes), associated with the **inflammatory response**, emerged as a biologically relevant candidate, given the well-established relationship between ageing and chronic, low-grade inflammation.

FEATURE SELECTION : CONSISTENT GENES

Pursuing the fight against batch effect

To mitigate potential unit-specific biases, we ultimately **retained genes whose correlation with age was consistent across care units**, aiming to reduce the influence of unit-dependent effects by prioritizing genes that exhibit a stable association with age across different settings. However, this selection strategy has important **limitations**: it is based on statistical **consistency rather than biological relevance** and **may therefore exclude genes with genuine age-related effects that vary across populations or clinical settings**. Given the limited number of care units available, the **estimated consistency may not generalize** to different settings.



ML STRATEGY

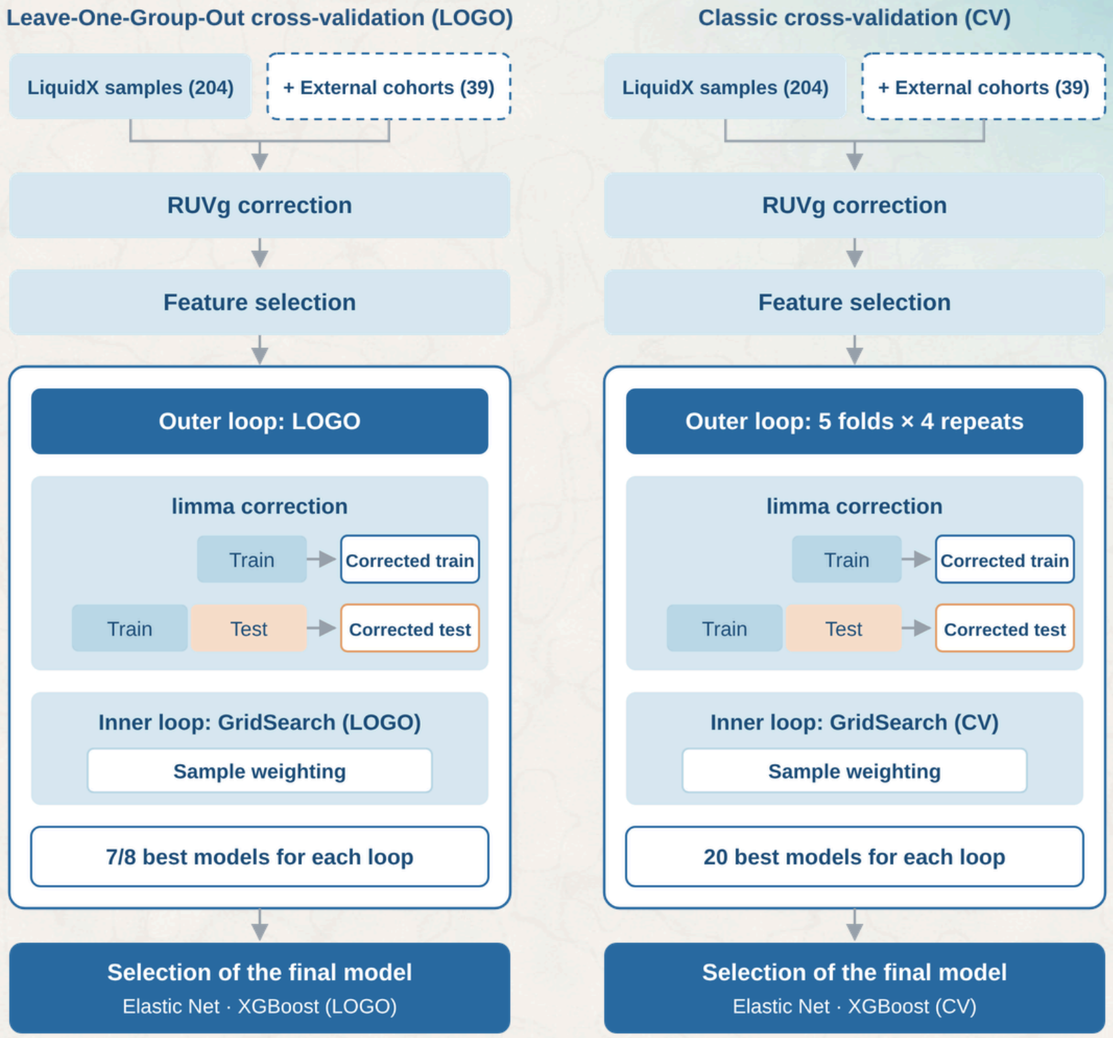
Classic CV against LOGO

Classic CV : patients are split at random into 5 folds, repeated 4 times, so 20 train/test splits. Each test fold mixes patients from every hospital.
LOGO : each fold holds out one entire collection unit. LOGO measures whether it generalises to a new centre.

We **weight the samples** because the models tended to predict values close to the cohort's mean age. Each patient gets a weight that grows with the distance between their age and the mean age.

We tried **three feature selection** explained before : Consistent genes, lasso genes (for XGBoost only) and co-expression modules.

We added public samples sequenced with a protocol similar to Flomics', to increase the size of the cohort.



SYNTHESIS

This internship started from an exciting idea: could a single blood draw reveal how fast our body is really ageing? The goal was to build a model that captures ageing and stays interpretable, so that beyond a number, it tells us something about the biology behind it. Every step pushed in that direction. We made sure the signal reflects ageing rather than how samples were collected, linked genes to real biological functions, and designed an evaluation that leads the model to learn what all patients share as they age.

And the promise of this work is worth it: help moving medicine from treating diseases once they appear toward preventing them.